

Hands-On 9 Project: HO9_NgRx_State

Commands:

```
ng new HO9_NgRx_State --routing --style=css
cd HO9_NgRx_State
npm install @ngrx/store @ngrx/effects @ngrx/entity @ngrx/store-devtools
mkdir src\app\store
mkdir src\app\store\course
ng g c pages/course-list
ng g c components/course-card
ng g s services/course
ng serve
```

What we did:

Implemented NgRx actions, reducer, selectors, store registration, and course list using store.

course.actions.ts

```
import { createAction, props } from '@ngrx/store';

export const loadCourses = createAction('[Course] Load Courses');
```

```
export const loadCoursesSuccess = createAction(
  '[Course] Load Courses Success',
  props<{ courses: any[] }>()
);
```

```
export const loadCoursesFailure = createAction(
  '[Course] Load Courses Failure',
  props<{ error: string }>()
);
```

course.reducer.ts

```
import { createReducer, on } from '@ngrx/store';
import { loadCourses, loadCoursesSuccess, loadCoursesFailure } from './course.actions';
```

```
export interface CourseState {
  courses: any[];
  loading: boolean;
  error: string | null;
}
```

```
export const initialState: CourseState = {
  courses: [],
  loading: false,
  error: null
};
```

```
export const courseReducer = createReducer(
  initialState,
  on(loadCourses, state => ({ ...state, loading: true })),
  on(loadCoursesSuccess, (state, { courses }) => ({
    ...state,
    courses,
    loading: false,
    error: null
  })),
  on(loadCoursesFailure, (state, { error }) => ({
    ...state,
    loading: false,
    error
  })))
);
```

course.selectors.ts

```
import { createFeatureSelector, createSelector } from '@ngrx/store';
import { CourseState } from './course.reducer';
```

```
export const selectCourseState = createFeatureSelector<CourseState>('course');
```

```
export const selectAllCourses = createSelector(
  selectCourseState,
  state => state.courses
);
```

```
export const selectCoursesLoading = createSelector(
  selectCourseState,
  state => state.loading
);
```

```
export const selectCoursesError = createSelector(
  selectCourseState,
  state => state.error
);
```

app.config.ts

```
import { ApplicationConfig, provideBrowserGlobalErrorListeners, provideZoneChangeDetection }
from '@angular/core';
import { provideRouter } from '@angular/router';
import { provideStore } from '@ngrx/store';
import { provideStoreDevtools } from '@ngrx/store-devtools';
import { routes } from './app.routes';
import { courseReducer } from './store/course/course.reducer';
```

```
export const appConfig: ApplicationConfig = {
  providers: [
```

```

    provideBrowserGlobalErrorListeners(),
    provideZoneChangeDetection({ eventCoalescing: true }),
    provideRouter(routes),
    provideStore({ course: courseReducer }),
    provideStoreDevtools({ maxAge: 25 })
  ]
};

```

course-list.ts

```

import { Component, OnInit } from '@angular/core';
import { CommonModule } from '@angular/common';
import { Store } from '@ngrx/store';
import { Observable } from 'rxjs';
import { CourseCard } from '../components/course-card/course-card';
import { loadCoursesSuccess } from '../store/course/course.actions';
import { selectAllCourses } from '../store/course/course.selectors';

```

```

@Component({
  selector: 'app-course-list',
  standalone: true,
  imports: [CommonModule, CourseCard],
  templateUrl: './course-list.html',
  styleUrls: ['./course-list.css']
})
export class CourseList implements OnInit {
  courses$: Observable<any[]>;

```

```

  constructor(private store: Store) {}

```

```

  ngOnInit() {
    const courses = [
      { id: 1, name: 'Angular', code: 'ANG101', credits: 4 },
      { id: 2, name: 'React', code: 'REA101', credits: 3 },
      { id: 3, name: 'NodeJS', code: 'NOD101', credits: 4 }
    ];

```

```

    this.store.dispatch(loadCoursesSuccess({ courses }));
    this.courses$ = this.store.select(selectAllCourses);
  }
}

```

course-list.html

```

<h2>Course List Using NgRx Store</h2>

```

```

<app-course-card
  *ngFor="let c of courses$ | async"
  [course]="c">
</app-course-card>

```

Output screenshot: Course List Using NgRx Store and Redux DevTools showing course state.

EF Core 8.0 LabStudentCoursePortal

Ask Gemini

localhost:4200

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Student Course Portal

This portal helps students browse courses, enroll in subjects, and view their profile.

Dashboard

Courses Available: 12

Enrolled: 3

GPA: 3.8

Enroll Now

Search courses

Searching for:

Students from REST API

ID	Name	Email
1	Leanne Graham	Sincere@april.biz
2	Ervin Howell	Shanna@melissa.tv
3	Clementine Bauch	Nathan@yesenia.net
4	Patricia Lebsack	Julianne.OConner@kory.org
5	Chelsey Dietrich	Lucio_Hettinger@annie.ca
6	Mrs. Dennis Schulist	Karley_Dach@jasper.info
7	Kurtis Weissnat	Telly.Hoeger@billy.biz